

'If It Doesn't Work for Me as a Researcher, Let Alone Others': Understanding Wellbeing Through an Autoethnography of Wearable and Non-Wearable Technologies

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Abstract

Wellbeing plays a crucial role in our daily lives, yet achieving or maintaining it remains a complex phenomenon, even for researchers. Capturing wellbeing data through wearable and non-wearable technologies to understand the state of “feeling good” poses significant challenges, requiring careful consideration of various factors. This autoethnographic study reflects on personal experiences with wellbeing data to explore the possibilities and limitations of these technologies. Within the constraints of limited time, resources, and circumstances, this study aims to 1) Investigate the potential and challenges of using wearable and non-wearable technologies for managing wellbeing; 2) Derive lessons from personal reflections to understand how multimodal data can provide meaningful insights and reshape the interpretation of wellbeing. The findings provide valuable insights into the integration of wearable and non-wearable data and offer directions for future research, particularly in designing technologies that enhance the understanding and quality of wellbeing.

CCS Concepts

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**.

Keywords

Autoethnography, wearable technology, non-wearable technology

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1 Introduction

Wearable technologies are readily available on the commercial shelf for anyone anticipating ownership. The proliferation of these commercial devices has enriched our lives by providing insights into our active and inactive everyday states. A wealth of information is at our fingertips for those interested in self-awareness. Often, such devices are used for everyday fitness, health, wellbeing tracking, and more, especially for wellbeing management to those with health-related issues. One example is COVID-19, where researchers utilize and self-studied wearables to gain insights into the lifelong impact of the virus and its potential applications in the design of pacing technologies [18]. Despite the effective use of devices in various contexts, understanding the use of wearables requires further understanding, as individuals may misinterpret and subsequently mistrust the represented data, leading to reluctance in using these devices. An understanding of prospective data and its significance from devices should form a foundational awareness, enhancing one's ability to derive meaning from their personal data.

Wellbeing data gathered from non-wearable and wearable devices offer several benefits. For instance, individuals can access multimodal data, ranging from physical activities to sleep qualities, and track essential health indicators such as electrocardiogram (ECG or heart rate), blood oxygen level, and wrist temperature [22]. This rich data can provide insights into their activity levels, helping them improve their fitness and adopt a healthier lifestyle. However, wellbeing is a complex concept encompassing physical, biological, and psychological aspects [5]. Representing numeric data in graphs and diagrams can be challenging without a fundamental understanding of the underlying information, context, environment, and individual circumstances. Wellbeing data may become troublesome or even induce panic if not accompanied by such understanding [2]. Despite the potential for gaining insights into wellbeing through wearable and non-wearable technologies, there is a noticeable absence of substantial discussion regarding how individuals perceive the combination of data from these sources and the significant role that a user's context plays in deriving meaning from this information. Wearables and non-wearables, with companion applications, enable users to monitor physical activities like steps, running, walking, and sleep quality/quantity. While these applications generally align with actual activities, occasional discrepancies may occur. Users can adjust and correct mistakes, but this process might be unclear, leading to potential frustration, as observed in a previous study [2]. Although wearables offer rich information for reflecting on wellbeing, their significance is amplified when combined

with a diary study. This reflection allows individuals to connect daily experiences with quantitative data, enhancing self reflection. By combining wearable data, a diary study, and autoethnographic insights, researchers can gain a comprehensive understanding of devices, context, and potential uses. This dual perspective, both as a researcher and in everyday use, shapes the design pathway for integrating technology into daily life.

This study involves the setup and execution of an autoethnographic study as a means of understanding, reflecting on, and rethinking the use of non/wearable technologies in everyday life. I discuss examples of ethnographic and design practices and present my approach to achieving a deep understanding, providing direction for further use in the study of understanding wellbeing and the self, and sharing lessons learned from the undertaking. The sensing capabilities of wearables allow researchers to collect multiple data points related to physiological phenomena and gain a deeper insight based on individual perspectives, contexts, environments, and social engagements with wellbeing.

2 Related work

Wearables and non-wearables. Wearable technologies have become more affordable [10], enabling most people to purchase them for monitoring their health and wellbeing [27]. Well-known companies in health and fitness domains, such as Fitbit, Apple, and Polar, consistently design and upgrade advanced sensing capabilities in the rapidly growing market [16]. The trend of using such technologies is increasing [37], as they are user-friendly [32] and enable individuals to monitor their active lifestyle, including fitness activities [1] or sleep quality [23, 43]. Commercial-grade or research-grade devices provide potential benefits and opportunities for almost everyone [43]. For example, some patients, caregivers, and healthcare professionals perceive the value of real-time feedback and unobtrusiveness when using them [4, 33]. Wearables are not only used in physical activities but also in the domains of mental health [17], chronic disease [39], tracking COVID-19 infections [14], and for general tracking [37]. Non-wearable technologies, or contactless technologies have become convenient and require minimal contact for individuals. The absence of visible devices may lead to minimal psychological and physiological challenges [46, 51]. They can be used for remote monitoring of body functions through embedded sensors, and doctors are increasingly interested in the use of such sensing technologies [44]. Examples of contactless technologies related to sleep quantity and quality matrices include Emfit QS, Beddit, and Withings. Typically, these technologies are used for sleep measurement and evaluation [3]. Consequently, there are valuable opportunities to explore contactless technology in the areas of health and wellbeing, particularly sleep. Most studies [3, 7, 19, 20, 42, 44] in the field of contactless technologies have focused predominantly on objective measurement, while subjective measurement could provide valuable insights for further deployment. In this work, we will refer to these technologies as non-wearables.

Autoethnography. In the realm of autoethnography, a dual role is assumed, acting as both a product and a process. In this context, the researcher's own experiences are positioned as the central data to be comprehended [13]. The incorporation of embodied experiences

in autoethnography, as described by Ellingson [12], directs attention not only to what is said or done but also to how it is felt, and how the researcher's body is positioned and understood in space. As a researcher is an expert in the area of their particular topic, they provide constructive knowledge on what design should be or should not be; and this is critically important when embarking on the research participant in a small-scale study or even a large-scale study. Providing examples of autoethnographic works, specifically addressing self-tracking of long covid [18], using a wearable to monitor blood pressure [36], or even living with drones for child-drone interaction [15], using artifacts for sleep ecologies with undergrad students [30], and living without mobiles [31], the commonality among these research studies provides a guideline and recommendation for future designs in these specific technologies. The studies often range from 2 weeks to years. They involve the setup and execution of autoethnographic studies as a means of understanding, reflecting, and rethinking how technologies are used in everyday life, without strict deadlines for researchers or prospective participants. I employ a similar approach of ethnographic and design practice examples, presenting our methodology to convey a deep understanding, provide pathways for further use in the study of understanding wellbeing and sharing lessons learned from it.

As a core element of autoethnographic study, journals or diaries play an important role in capturing behaviors, experiences, and feelings. Several studies have also utilized diary studies to demystify potential benefits and pitfalls in exploring the use of wearables in the context of health and wellbeing. For example, Ambe et al. employed a diary for older adults' wellbeing to reveal different perspectives on how data is generated, represented, and shared among aging couples, as well as how parents perceive the use of baby wearable technology [50], and how children with autism use mobile applications [41]. Pennebaker and Smyth also focused on issues related to physical and mental health [38], while Kong et al. sought to understand the relationship between mobile use and young adults [26]. Furthermore, a recent study on Covid-19 behavior was investigated using a daily diary study [48]. Turner et al. also investigated the use of a diary study in home security. This study unpacks the impact of using both wearables and non-wearables through an autoethnographic study on a broader range of personal situations and social connections [49].

3 Methodology

3.1 Data collection and analysis

3.1.1 Data Collection. An Apple Watch Series 8 (version 10.2) was employed for collecting physiological and sleep data, with data transmission to the HealthKit app for my wellbeing overview. My focus encompassed heart rate, heart rate variability, blood oxygen levels, and physical activities. The latest Apple Watch version introduced sleep tracking, enabling us to access sleep quantity data. Data synchronization occurs through the iPhone, accessible via the HealthKit Dashboard. Post a June 2023 operating system upgrade for the iPad HealthKit application¹, I now have an expanded dashboard for a more comprehensive view of health and wellbeing data, facilitating easier visualization of graphs and data. In addition

¹<https://www.apple.com/au/newsroom/2023/06/apple-provides-powerful-insights-into-new-areas-of-health/>

to wearing the watch, I incorporated the Withings smart mattress into my bed to complement and track sleep quantity and quality. An emerging concept in non-wearable devices is represented by the Withings Sleep Analyzer (WSA)², which can be placed under mattresses in a non-intrusive manner. Powered by a deep-learning algorithm, the WSA utilizes data on body movement, breathing patterns, cardiac activity, and snoring to calculate an apnea-hypopnea index (AHI), indicating the level of sleep apnea. The WSA is a clinical-grade device, evaluated for validity against the gold standard of sleep monitoring [11], requiring minimal installation and connecting via the mobile app called Withings app. It only requires a one-time setup, with no need for any further installation. The data is typically ready on the iPhone every morning once I wake up. A diary study is employed as a complement to the data I obtain from wearable and non-wearable devices. This approach allows us to understand why things happen the way they do and how individuals comprehend my own data. Several prompts are provided for the diary to gather information related to wellbeing data, including biological, psychological, and social aspects. In addition to pre-prompt questions, there is room for freestyle ideas and thoughts to be written. The auto-ethnography-inspired self-study lasted for a continuous 4-month period, reflecting actual day-to-day experiences. Despite my efforts to consistently maintain the diary study, there were occasions when I was not unable to make an entry, recognizing the realities and unpredictabilities of daily life. While I attempted to record perceptions through video recording, it was not done effectively. However, completing a pen and paper diary was feasible. As someone with an IT background, working with typing and an online diary was convenient and could be done anywhere. The author is aware of this limitation and recognizes that this online platform may not be suitable for everyone. However, the diary study is designed for easy printing and distribution, facilitating its use in future research, particularly with a focus on studies involving aging individuals. A sample diary is in appendix B adapted from Ambe et al..

3.1.2 Data analysis. Data from the Apple Watch and smart mattress were collected via applications and analyzed in CSV or text formats, generating millions of data points from both wearable and non-wearable devices. Over 122 days (approximately four months), I also maintained my wellbeing diary, using prompts to reflect on daily life and identify areas for improvement. These reflections, combined with the data, provided evidence-based insights into my wellbeing. While applications offered graphical dashboards for routine monitoring, episodes of compromised wellbeing prompted deeper analysis. I visualized the data using R to examine specific periods and combined it with detailed diary entries to gain meaningful insights (Figure 1).

3.2 Situational overview

During a challenging period, I encountered health issues such as over fatigue and food poisoning, leading to a serious incident where I fainted in my apartment and hit my head on the tiled floor. Fortunately, I regained consciousness after a few minutes, managed to get to my bed, and assessed my condition. Living alone, this

experience was a wake-up call for myself to pay closer attention to my health and wellbeing. Realising the potential of the devices I had, both wearable and non-wearable, I recognised how they could be instrumental in monitoring and managing my health. This incident, coupled with the study, expanded my understanding of how wearables can not only track health issues but also facilitate a shift towards a healthier lifestyle. The study was conducted over 4 months for myself to explore the possibility of using both wearable and non-wearable devices for an early-middle age adult[34]. Before engaging with other user groups (e.g., late middle-aged adults, older adults, and/or people with intellectual disability), I conducted a few studies based on my experiments. Since wearable and mobile computing technologies have become widely available and accessible, it is important to understand user experiences, particularly with someone skilled in troubleshooting these technologies. I aimed to investigate how these experiences shape one's management of wellbeing and how usage could be improved to effectively address any unforeseen issues.

4 Findings

4.1 Complement approach for self-reflection

Similar to [49], I was initially concerned about not being able to complete a diary entry every day, as the objective was to collect as much data as possible. However, in reality, certain days posed challenges due to health, wellbeing, and work commitments, leading to some days not being documented in the diary. Nevertheless, the data collected offers a genuine reflection of real-life experiences. Consequently, the diary entries varied in length, ranging from 100 to 500 words, and included diverse content such as pictures of food, graphs, numerical data, images of skin rashes, and screenshots of various applications. In conjunction with diary entries, objective measurements were obtained from Apple's Health Kit app and Withings's app almost every day, except when not wearing or being nearby the devices. Reports from these applications helped myself to reflect on the day's events and whether they aligned with my perceptions. I observed that while daily entries could facilitate routine data collection, periodic reflections—such as on a weekly basis—would provide a broader perspective on overall feelings. In some instances, I recorded entries in the app upon awakening in the early morning to document reflections on sleep. However, mornings are often rushed with work and other matters, resulting in only minimal entries with a few words (10-20) being logged.

Diary prompt questions and freestyle writing enhanced my experience with data gathering from the applications. For example, *"I think I know, but sometimes additional information can help me to understand more before sharing with others"*. A mixed method of gathering data proves valuable at certain times or occasionally. I often refer back to the diary to check what happened during a particular day of interest and reflect on how well the graphs capture daily wellbeing. A daily diary proved to be effective and important especially in sleep [24] which may overcome the limitation of objective measurements. With three main data gathering approaches (Health Kit data, Withings data, and diary) complemented by self-reflection, I found that collected data complementing each other to comprehend my wellbeing. A few night that I forgot wearing my watch and I noted my diary that: *"Because I didn't wear the watch*

²<https://www.withings.com/be/en/sleep-analyzer>

last night, I only viewed the data from the Withings app. Actually, the Withings app is not too bad compared to HealthKit. HealthKit is a bit hard to read, and it's challenging to absorb information. — 29/11/23 Diary”.

Challenges in finding moments for self-reflection arise on days when I could not wear my smartwatch (e.g., due to skin rash), sharing the bed with my sister when she had a short visit (e.g., causing the smart mattress to be unpromising), or forgetting and feeling overwhelmed (e.g., a daily diary may not be feasible). Reflection typically occurs in response to anomalies or health issues, utilizing all available data in the specific context. Real-time reflection may not always happen during challenging moments, but the combination of objective measurements and subjective observations contributes to constructing a narrative emphasizing the significance of self-care. This is exemplified in instances like a fainting episode due to food poisoning, discussed later in this paper.

4.2 Unpacking importance: a closer look at wearables and non-wearables

Comfort. The selection of wearables should be a careful consideration, weighing both positive and negative aspects [10]. This becomes especially crucial when the study aims to explore sleep behavior and physical activities over extended periods, such as weeks, months, or years. The continuous 24/7 wearing of a smartwatch may introduce discomfort, leading to a potential decline in engagement with such technologies. Salim’s study also observed that participants experienced discomfort from smartwatches, leading them to take them off [45]. Although I typically wear the smartwatch during the day without issue since the introduction of the first generation by Apple, wearing it throughout the night for sleep exploration revealed discomfort. A similar view observed in active wearers as reported by Jeong and team [25]. Factors such as humidity, environment, and the material of the band contributed to skin rashes, prompting intermittent data recording on some days. A subsequent change was made to a more comfortable nylon band known for its soft cushioning and breathability.

Complementarity. The notion of ‘Less is more’ prompts the question of how much less is optimal. Rather than introducing multiple data collection devices simultaneously, I started with a single wearable, adding a non-wearable to complement data collection at a later stage. The smartwatch, requiring charging every day or two, faces challenges such as forgotten wear or unexpected circumstances, prompting consideration of non-wearable alternatives. Different applications were used for monitoring heart rate and sleep data, including clinical validated non-wearable devices. While non-wearables are effective for solo sleep monitoring, challenges arose when my sister visited, affecting data reliability. During this period, wearable devices became the preferred, more accurate source for monitoring sleep and related metrics.

Engagement. Social engagement is a crucial consideration, especially during prolonged device usage. Aligning with Menheere’s study [35], which emphasizes user empowerment through social support for exercising, I discovered a compelling aspect: sharing physical activities with a family member overseas. Living alone, away from family, this interaction with data proved significant for me. Each time I exercised, capturing calories burned and distance

covered, I shared my wellbeing information with my sister. This reciprocal communication motivated both to exercise more and challenge each other, exchanging pictures of walking distances and exercise burn. These exchanges go beyond data, offering a glimpse into life, promoting a sense of presence and connectedness [21, 29], acting as social support despite geographical and temporal distances with family.

5 Study implications and Design considerations

5.1 Pathways to bridge the data gaps between technology and individual

From 2017 to 2023, I accumulated millions of general activity records and some vital signs for personal data, regularly viewing the app for day-to-day observations. However, my interaction was limited to a brief glance at graphs, sharing exercise results through pictures with family and friends, showcasing walking distances. This pattern, akin to daily wearing habits [25], changed as I delved into technology understanding for studies involving diverse demographics. Initially, my focus was on technology utility, fostering a healthier lifestyle. Integrating existing technologies significantly broadened my understanding of multifaceted wellbeing, particularly when combined with a diary study, as demonstrated in this paper and our prior work [2]. Quantitative data from wearables and non-wearables monitor my physical activity levels and optimal sleep duration through numerical and graphical representations. Importantly, I encountered health issues that empowered me to leverage the data for personal and collective benefits, aligning with the study of personal health using fitness tracking technologies [18]. I often asserts, “If it does not work for me as a researcher, let alone others.”. Upon reflecting on the obtained data, I realized that the meaning and awareness of data stem from multiple factors — experiences with numeric data, using my reflections to transform it into information, and ultimately knowledge.

5.2 Technologies as a complement and social support

Wearable and non-wearable technologies dominate the market. I use my smartwatch everyday, assisting in work management through wrist reminders and stand alerts. However, I often overlook the prompt to stand due to work commitments or situations where standing is impractical, echoing findings from Jeong and team [25]. In extreme cases, like being on a plane or in a meeting, responding to such alerts can be challenging, reflecting shared user experiences. Wearable devices contribute to a unique user narrative based on their design and functionalities, offering distinct experiences through embedded sensing capabilities like haptic feedback and close physical proximity. This experience is shaped by past technologies and introduces novel modes of interaction, potentially encountering user resistance [52]. During a period of food poisoning that left myself dizzy and on the floor, the absence of any devices prompted me to reflect, wishing I had my smartwatch to check my heart rate. The question popped into my mind: “Where was my smartwatch? I wanted to wear it and check if my heart rate was all right”. Later, when I recovered, I realized that “when I needed the wearable the most, it wasn’t there to even help”. This critical

situation raised concerns for my family and friends. Subsequently, a wearable became a primary item for me to wear it almost all the time.

I later shared my difficult time with my friend, who was subsequently worried about myself. I discussed and decided that my friend would check if I was alright throughout the day at random points in time without compromising my privacy. I agreed that text messages would be an appropriate method for that period. This is particularly important to consider on top of the capacity of technology when it comes to sharing or caring for each other. What information should I share with others? Whom should I allow to see that information, and for how long should I allow them to see it?. Social support is crucial for the functioning of our society, as I live in communities. Lee and Coughlin not only suggest that social support helps older adults overcome barriers in technology adoption but also empowers myself, as a mature adult, to feel safe and connected with my close circle, facilitating the sharing of information through technology.

5.3 Strategizing a number of devices needed

Determining the optimal number of devices involves multiple experiments and careful consideration, weighing benefits and drawbacks in a given time and context. Devices should be user-friendly with easy application setup, especially in an unobtrusive manner. I initially wore a smartwatch continuously day and night, occasionally needing to push myself to do so, with variable success. Recognizing the need for complementary methods, I later experimented with smart mattresses in conjunction with the smartwatch. These mattresses required minimal setup—simply going to bed provided readily available sleep data for review upon waking. Managing a smartwatch involves constant reminders for myself to charge and wear it before bedtime. Choosing between non-wearable and wearable devices presents trade-offs, making the decision challenging. However, having both can complement the data collection process. For instance, if the smartwatch is forgotten, data from other devices can still be relied upon. While not a new approach, several studies [6, 8, 9, 44] utilize multiple devices to gather extensive data for broader analyses and understanding individuals on a larger scale.

5.4 Data representation matters

Data is vital for conveying meaningful knowledge to individuals [2]. Although I used a smartwatch since its launch in 2015, I did not explore data details in design or development. Accepting numbers without further exploration was common, reflecting a trend among users who integrate technology seamlessly into their daily lives [25, 40, 47]. Based on my reflection, there are several design recommendations that researchers should consider when expanding this study:

- **Adaptive device pairing:** Previously, I did not critically evaluate my health conditions using the app efficiently. Close examination of two devices and applications revealed interesting circumstances where data could be used interchangeably. If one device is missing, I can use another to understand wellbeing conditions. My primary interest lies in obtaining a summary of overall activities and sleep, achievable through complementary devices in the study.

- **Minimalist data visualization:** To understand the overall situation (appendix A), I figured out some baseline of normal blood oxygen level or heart rate while I'm awake or asleep. This part, I found, is an important anchor to further into specific circumstances or attain curiosity. The baseline should stand out and easily absorb information, although most applications provide some baseline information. However, I found it is challenging if it requires effort for the user to navigate to other pages. This can cause a loss in navigation and obtaining information for a particular interest. The baseline and 'your' data should stay within the same view to allow individuals to compare and learn whether the situation is a concern or of interest. I often struggled to understand acceptable blood oxygen levels, highlighting the need for readily available standard information alongside graphs and numbers. It's crucial to minimize information to help individuals comprehend their current situation without overload. Establishing a standard for physiological data allows for meaningful reflection. The focus shifted to assessing data reliability at specific moments, such as a fainting episode. Despite the app providing individual vital sign graphs, comparing different vital signs remains challenging.

- **Facilitate Users in connecting data with knowledge:** HealthKit data was exported for analysis, focusing my particular interest in the fainting situation, specifically blood oxygen level and heart rate. A potential correlation between low blood oxygen and elevated heart rate was explored, as shown in Figure 1a and 1b. I recalled that rapid breathing during shortness of breath can lead to increased heart rate, a process controlled by the respiratory center in the brain. My insights were further complemented by online resources from Webmd³ and Mayo⁴. I obtained health knowledge, leading to an awareness of healthy living. This insight is shared with friends, family, colleagues, and potential project participants [2]. The study's wellbeing data encourages reflection on personal activities, fostering insightful information generation and empowering critical evaluation. These experiences and insights are shared within a trusted circle, promoting companionship in one's wellbeing journey. These insights guide engagement strategies with future participants from various target groups.

6 Limitations and future work

This study integrates autoethnography, multimodal data collection, and diary studies to offer a comprehensive perspective on wellbeing management. It uncovers gaps in data representation, highlights usability challenges, and brings together technical evaluation with emotional and social aspects of data use. However, there are some limitations to consider, including the reliance on a single participant and the complexities of data sharing, which may impact generalizability and user privacy. While the autoethnographic lens provides rich introspection, this study has limitations, particularly due to the use of a single participant, which may lead to single-user bias

³<https://www.webmd.com/lung/pulse-oximetry-test>

⁴<https://www.mayoclinic.org/diseases-conditions/tachycardia/symptoms-causes/syc-20355127>

and limits generalizability. To enhance the study, aligning with embodied ethnography frameworks could provide a broader perspective by examining how other users experience both wearable and non-wearable technologies through their bodies and interactions, rather than relying solely on self-reports. Additionally, since the findings are based on my personal experience, including multiple participants or comparing insights across different ethnographic frameworks would help reduce bias and provide a more comprehensive understanding of technologies. Although this method does not aim for generalization and may not be able to achieve it, the findings remain valuable to the HCI community. Another aspect I should emphasize is that data sharing among friends or family can pose potential privacy and ethical concerns. While data sharing can enhance social engagement and awareness among close ones, future studies should focus on how data owners are willing to share their data and their ability to control it.

7 Conclusion

Wearable and non-wearable devices offer valuable capabilities for managing wellbeing data, each with its benefits and pitfalls due to compatibility issues and varying effects on individuals. I employed a qualitative approach, utilizing self-reflection and collaborative discussions of my hands on experiences with smart devices, aiming to understand their impact through a diary study. This study explores the significance of both types of devices, emphasizing aspects of comfort, complementarity, and engagement. Combining quantitative data on vital signs, physical activities, and sleep with the diary study addressing gaps in unknown circumstances. My approach underscores the importance of wellbeing data, providing meaningful information for individuals, close relationships, and potential study participants. Wellbeing data empower individuals to reflect, understand, and manage their experiences, fostering insightful knowledge. The key is empowering individuals to learn through reflection on their own data, bridging the gaps between tracking technologies and their users as data owners. These insights can be shared with friends, family, and prospective participants, following the trajectory of Data-> Reflection-> Meaningful information-> Ability to learn-> Transform to awareness -> Share with others.

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8 Appendices

A A sample of one entry of diaries

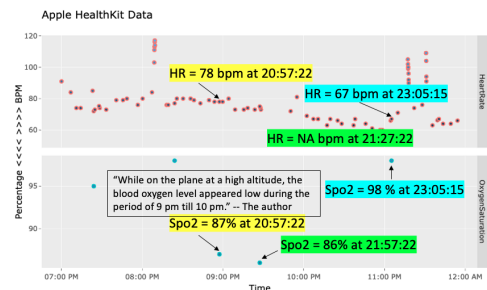
"I checked my device yesterday (26/11/23) for my blood oxygen level, and it was a perfect 100%. I'm curious if that's actually the real reading because it seems really good. Now, after searching, I see that 95% to 100% is considered normal. If it's below 95%, one should seek advice from a doctor (information obtained from the internet). Looking at my sleep data, I think I only slept for 5 hours and 54 minutes last night.

Not so good because I slept late, and today is a working day. My RR (sleep respiratory rate) seems to be normal (healthy adults ranging from 12-20 breaths per minute during sleep). For HRV, today's number is 35 ms, which seems to be low (I'm not sure if this number is during the night or during the day, but it looks like the standard normal HRV should be around 100 ms—now I'm worried!). I searched for information about it, and it also depends on your age and fitness level. (See more information on the internet; SDNN seems to be used in medical settings to understand the level of HRV). – 27/11/23 Diary"

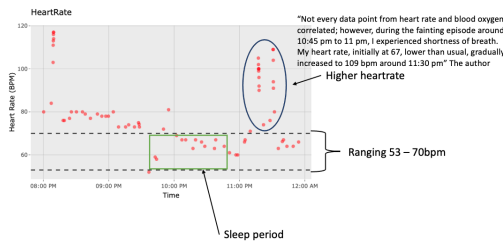
B Diary Questions

- Overall feeling of the day
- What interesting happened today (or what made today a good/or not so good day)?
- What makes you curious of the data collected by the devices today? What do you think of the information it shows? Any reflection of the device information?
- Additional notes: Any comments or thoughts about today, and the experience with the devices in your home?
- My Biological data entry (Physical health, genetic, disability etc)
- My Psychological data entry (Emotion, Coping skills, attitude & belief)
- My Social data entry (education, social support, peer relationship, socio-economic background)
- Reflection and free style thought

C Figures



(a) A graph shows the heart rate and blood oxygen level generated using R studio. Heart rate data was sampled a lot while blood oxygen level has minimal datapoint, therefore, it was challenging to provide meaningful conclusion at this stage apart of some information here and there.



(b) A graph shows the heart rate at a closer look. There was a period that I was asleep before experiencing fainting in the green rectangle box, my ranging sleep heart rate was between 53 - 70 bpm. After awakening to experienced dizziness and short breath, my heart rate was gradually increased, highlight in oval.

Figure 1: A graph generated by R Program to vision the closer look of the data.

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